Ben Stan

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Selected Work Experience

Catalysis & Nanomaterials Research Assistant
Klinkova Lab, University of Waterloo | May–Aug 2025
- Optimized decoupled electrolysis using NiOOH nanoparticle electrodes and electro-Fenton oxidation to improve efficiency and catalyst stability.
- Designed and fabricated carbon-paper electrodes (carbon-black catalytic inks) with enhanced activity and conductivity.
- Performed electrochemical testing and troubleshooting to improve system reliability and reproducibility.
- Supported catalyst preparation and nanoparticle surface modification for performance enhancement.

Process Engineering Co-op
AirBoss of America | Jan–Apr 2025
- Optimized mill temperatures and profiling to identify hotspots and improve rubber-processing stability.
- Evaluated re-milling impacts and defined optimal reprocessing conditions for consistent material performance.
- Automated workflows (scheduling, reporting, data organization) using Python, reducing manual workload.
- Supported high-throughput process troubleshooting and continuous improvement.

Chemical R&D Lab Assistant
Woodbridge | Jan–Apr 2024
- Investigated TPU behavior and processing, supporting experimental design and lab-scale trials.
- Conducted LCA and PCF assessments to guide sustainable material decisions.
- Built Power BI + Power Automate systems for automated scheduling, reporting, and workflow optimization.
- Designed SharePoint interfaces for centralized data access and documentation.
- Assisted with patent development, including literature review and technical writing.

Lab Assistant
University of Waterloo | Sep–Dec 2021
- Operated and troubleshot distillation, LLE, centrifugation, and GC systems to improve reliability.
- Optimized a semi-batch ethanol bioreactor, improving operating conditions and data consistency.

AI Research Assistant
University of Waterloo | Feb–May 2021
- Recreated and validated ML models from research papers, improving architectures and reproducibility.
- Trained and analyzed image-recognition models to guide model selection and performance tuning..

Technical Skills

Electrochemistry, Nanoparticles, Catalysis, Surface Engineering, Process Intensification, GC, IC, NMR, UV-Vis, Bioreactors, Distillation, LLE, Gas Chromatography, Python, Power BI, Power Automate, SharePoint, Data Pipelines, SOP Development, Machine Learning (PyTorch), Process Design, Process Optimization, Process Flow Diagrams, Excel, Word, Process Simulation.

Highlights

-30 months of engineering co-op experience across industry, R&D, and academia
-Certifications in process intensification and plasma physics
-Winner, Canadian Nuclear Laboratories Innovation Challenge
-Doubled bioreactor performance.

Education

Bachelor of Applied Science, Chemical Engineering
University of Waterloo — Waterloo, ON Sep 2020 – Present 4A Term

Certifications

-AIChE's An Introduction to Large-Scale Energy Production
-RAPID's Fundamentals of Process Intensification
-RAPID's Intensified Reaction Processes
-IAEA's Fundamentals and Application of Plasma Physics including Fusion Plasmas
-AIChE's The Nuclear Fuel Cycle in the U.S.: What Are the Options?

Selected Projects

Nuclear Innovation Challenge
-Designed a security system for CANDU reactor spent fuel.
Capstone Project
-Designed a P&ID, in aspen plus to use the Hydrometallurgical process to recycle NMC batteries.
Heat transfer project
-Used finite difference methods to simulate heat transfer through a cup to the surrounding in.